Higher National Diploma in Engineering (Civil/ Electrical & Electronic/ Mechanical)

First Year, First Semester Examination — 2022

22 ENI 104-Program Concepts

Question 01 (Total 20 Marks)

i. Define the term 'software engineering'. (02 Marks)
ii. Briefly describe two (02) objectives of software engineering. (04 Marks)
iii. Answer the following questions considering the 'waterfall model'.
   a. Mention the type of requirements to select waterfall model for system development. (02 Marks)
   b. Draw a named diagram showing all the phases of waterfall model. (04 Marks)
   c. Describe any two (02) phases of waterfall model. (04 Marks)
   d. Mention two (02) disadvantages of waterfall model. (02 Marks)
iv. Name two (02) process models that you can use for Rapid Software Development (RAD). (02 Marks)

Question 02 (Total 20 Marks)

i. List four (04) requirements elicitation methods/ tools. (04 Marks)
ii. Briefly explain the difference between validation and testing in software engineering. (04 Marks)
iii. Briefly describe the following terms:
   a. Algorithm. (02 Marks)
   b. Flowchart. (02 Marks)
iv. Write an algorithm to find the maximum of two given values. (04 Marks)
v. Draw a flow chart to find the sum of all numbers from 1 to 10. (04 Marks)

Question 03 (Total 20 Marks)

i. What is a computer program? Give two (02) examples for programming languages. (03 Marks)
ii. Briefly explain the purpose of compiler / interpreter in programming. (03 Marks)

iii. Identify data types that you can use to store the following quantities: (04 Marks)

a. 45
b. 1.3147
c. D
d. True/False

iv. Mention whether the following variable names are valid or invalid: (04 Marks)

a. X
b. name
c. void
d. 2ndValue

v. Declare and initialize (assign values) variables or constants to hold the values given below. First one is done for you as an example. (06 Marks).

a. Saman (string x = "Saman";)
b. Value of pi (π)=3.14
c. -100
d. A

Question 04 (Total 20 Marks)

i. Using an example, briefly explain the importance of comments in programming. (04 Marks)

ii. Write the output of the programs given below: (08 Marks)

a.
```
#include <iostream>
using namespace std;
int main(){
 cout<<"HND\nEngineering";
return 0;
}
```

b.
```
#include <iostream>
using namespace std;
int main(){
 cout<<10%3*6/2-2*4<<endl;
return 0;
}
```

c.
```
#include <iostream>
using namespace std;
int main(){
 int x=5;
while (x<10){
 cout<<x<<endl;
 x++;
}
return 0;
}
```

d.
```
#include <iostream>
using namespace std;
int main (){
     int x=3;
switch(x){
     case 1:
     cout<<"High";
     break;
     case 2:
     cout<<"Moderate";
     break;
     case 3:
     cout<<"low";
     break;
     Default:
     cout<<"Invalid Input";}
     return 0;
}
```

iii. Write a computer program to read a number entered through the key board and print whether the number is odd or even. (04 Marks)

iv. Write a C++ function to accept two integer values and return the multiplication of them. (04 Marks)

Question 05 (Total 20 Marks)

i. Write the syntax of nested if else statement in C++. (02 Marks)

ii. Write a C++ program to find and display the grade of a subject when you enter marks. Consider the criteria given below to assign the grade: (08 Marks)

Marks	Grade
75 - 100	A
55 -74	B
40 - 54	C
0 - 39	F

iii. Briefly explain the purpose of an array variable. (02 Marks)

iv. Number of students available in different classes are given as: 40, 35, 34, 28 and 42. Write down C++ code to declare an array variable and initialize it with these values. (04 Marks)

v. Using a **for loop** write relevant codes to display all the values of the above array. (04 Marks)

Question 06 (Total 20 Marks)

i. What is a database? Give two (02) examples for DBMS software. (04 Marks)

ii. Write down one (01) advantage and one (01) disadvantage of DBMSs. (02 Marks)

iii. A canteen is selling food items to the customers. The food items are purchased from some suppliers daily. A database is used to store details about food items, suppliers and daily purchases. Three tables taken from the canteen database are given as follows. Consider them to answer the questions given below.

FoodItem

ItemID	ItemName	Stock
I01	Fish buns	15
I02	Tea buns	16
I03	Fish rolls	12
I04	Patties	23

SuppDetail

SuppID	SuppName	Phone
S01	Saman	0774523987
S02	Aamila	0783459876
S03	Nuwan	0715623908
S04	Kasun	0723454356

Purchase

SuppID	ItemID	PurchasedDate	Quantity
S01	I03	12/5/2023	45
S03	I03	13/5/2023	20
S02	I01	14/5/2023	60
S04	I02	14/5/2023	30

a. What is the purpose of having primary key in a table? (02 Marks)

b. Write down the table name and relevant primary key(s) for the three tables above. (04 Marks)

c. Mention suitable data types for the fields of Purchase table. (SuppID, ItemID, PurchasedDate and Quantity). (04 Marks)

iv. Briefly explain any two (02) of the following terms relevant to object-oriented design: (04 Marks)

a. Encapsulation

b. Abstraction

c. Inheritance